Finding stationary points by approximate Jacobian method

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In this document, I describe an algorithm to find stationary points of smooth vector fields, in the hope of applying it to nonconvex-nonconcave min-max problems. My motivation for taking the time to write up these notes is that this algorithm and its justification are so simple that I wonder why people never just do that.

1 An (oversimplified) overview of min-max optimization

In (nonconvex-nonconcave) min-max optimization problems, we are given a function $f : \mathbb{R}^{d_x} \times \mathbb{R}^{d_y} \to \mathbb{R}$ (or $f : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ in the constrained case) and we want to find a local saddle point of $\min_x \max_x f(x, y)$, i.e.,

$$\text{find } z^* = (x^*, y^*) \text{ such that } \forall (x, y) \text{ in a neighborhood of } z^*,\ f(x^*, y) \leq f(x^*, y^*) \leq f(x, y^*).$$

This problem is notoriously complicated. A slight relaxation, which is still complicated, is to ask only for a stationary point instead of a saddle point.

Pose the vector field $F(z) = \begin{pmatrix} \nabla_x f(z) \\ -\nabla_y f(z) \end{pmatrix}$ on $\mathbb{R}^d = \mathbb{R}^{d_x} \times \mathbb{R}^{d_y}$ (or on $\mathcal{Z} = \mathcal{X} \times \mathcal{Y} \subset \mathbb{R}^d$). In the unconstrained case, a stationary point of f is simply a z^* such that $F(z^*) = 0$. If the problem is constrained to a set $\mathcal{Z}$ and $\mathcal{Z}$ is convex – which we assume henceforth–, the natural notion of solution is given by the *Stampacchia variational inequality (SVI)*

$$\text{find } z^* \text{ such that } \forall z \in \mathcal{Z},\ \langle F(z^*), z - z^* \rangle \geq 0.$$

This variational inequality expresses that following the vector field $-F$ starting from z^* would point us to outside the set, or in technical terms, that $-F(z^*)$ belongs to the polar cone (the negative of the dual cone) of $\mathcal{Z}$ at z^*.

1.1 The usual view

Unsurprisingly given their equivalents for minimization problems, the three most common (deterministic) algorithms for min-max optimization are

- Gradient Descent-Ascent (GDA): $z^+ = z - \eta F(z)$. That is to say, fix $\eta > 0$, pick $z^0 \in \mathbb{R}^d$ arbitrarily and let $z^{k+1} = z^k - z^k - \eta F(z^k)$.
- A generalization of Extragradient (EG+): $z^+ = z - \lambda\eta F(z - \eta F(z))$ where $\lambda \in (0, 1]$ ("explore aggressively, update conservatively", to quote the title of a related paper).
- Optimistic Gradient Descent-Ascent (OGDA): a variant of EG which maintains two sequences of iterates and alternates updates between them.

Of course, the parameters η and λ can also be made to depend on k, or even be chosen adaptively depending on $F(z^k)$.

Just as convergence theorems for gradient descent-type algorithms exist for a variety of assumptions (convexity, lipschitz-ness, strong or weak convexity, smoothness, PL-inequality...), in min-max optimization a variety of conditions have been proposed which can guarantee convergence. Only the ones used in the specific recent works I reviewed are discussed below; for more, see e.g. [Böh22].

- Convex-concavity of f basically ensures that most of the intuition from minimization transfers to min-max optimization.
- We say F *satisfies the Minty condition* if there exists z^* which is a solution to the *Minty variational inequality*
$$\forall z \in \mathcal{Z},\ \langle F(z), z^* - z\rangle \leq 0.$$
It turns out that, under only smoothness assumptions, any such z^* is in fact a solution to the SVI. The converse is not true in general – unless f is assumed convex-concave. This is known as Minty's theorem.
- We say F *satisfies the ρ-weak Minty condition* if there exists z_0 such that
$$\forall z \in \mathcal{Z},\ \langle F(z), z_0 - z\rangle \leq \frac{\rho}{2}\left\|F(z)\right\|^2.$$
This condition was introduced (or maybe reintroduced, I don't know) recently by [DDJ21]. They show that, even though z_0 may not be a solution to the SVI, existence of such a z_0 implies existence of a solution to SVI when $\rho \leq \frac{1}{4\sup\|\nabla F\|}$ (later extended to $\rho \leq \frac{1}{2\sup\|\nabla F\|}$ by [PLPFC22]).

Recent works analyzing the convergence of EG+ under the weak Minty, or a related, assumption include [DDJ21], [PLPFC22], [Böh22], [HLG22].

1.2 An alternative approach

Gradient descent-ascent and its variants find stationary points based on the principle "just follow the vector field until you're stuck". [ALW19] propose a radically different perspective: actively try to minimize the magnitude of the vector field,[1] by running gradient descent on the objective
$$z \mapsto \frac{1}{2}\left\|F(z)\right\|^2.$$
They call this method Hamiltonian gradient descent, which I really don't like because "Hamiltonian" is also used to refer to a different object in optimization (even in the PhD thesis of one of the authors in fact). Interestingly, the same method is proposed in a remark in [Lu21], under the name Jacobian method. It's still an arbitrary name but it's slightly better, so that's the one I'll use in the sequel.

In the unconstrained case, the Jacobian method (JM) update is
$$z^+ = z - \eta\nabla\left[\frac{1}{2}\left\|F\right\|^2\right](z) = z - \eta\nabla F(z)\cdot F(z)$$
where $\nabla F(z)$ is the Jacobian matrix of F. It is also equal, up to skewing, to the Hessian of f: $\nabla F(z) = \begin{bmatrix}\partial^2_{xx}f(z) & -\partial^2_{xy}f(z)\\ \partial^2_{xy}f(z) & -\partial^2_{yy}f(z)\end{bmatrix}$. Note that since $\left\|F\right\|^2 = \left\|\nabla_x f\right\|^2 + \left\|\nabla_y f\right\|^2$, for JM it doesn't matter how we skew the vector field; we would get the exact same update if we used $\widetilde{F}(z) = \nabla f(z)$ instead of F. Importantly, JM is a second-order method in f, and its per-iteration cost is $O(d^2)$ if run naively, compared to $O(d)$ for GDA, EG+ and OGDA. Hessian-vector multiplications can be implemented efficiently though, so it's not that bad.

[1] It's a bit similar to how Newton's method can be motivated in 1D as finding the roots of f' instead of minimizing f directly.

Remark 1.1. [ALW19] and [Lu21] propose JM only for the unconstrained case. In the constrained case, if the solution of the SVI lies on the boundary of $\mathcal{Z}$, there is no reason for it to have zero gradient, and so there is no point in trying to minimize $\|F\|^2$. Still,

- In cases when the constraints defining $\mathcal{Z}$ are only (linear) equality constraints (since we assumed $\mathcal{Z}$ convex), and/or when we know that the solution lies in the relative interior of the feasible set, it is straightforward to adapt JM with the appropriate projections.
- In some inequality-constrained cases it is possible to modify JM to obtain approximate (in a sense to be determined) solutions of SVI. For example, one could try replacing the constraints by appropriate penalization terms in the min-max problem, or (equivalently?) adding to the JM update a term given by the gradient of a barrier function.
- One could also try to adapt JM by considering that the true objective to minimize should not be $\|F\|$, but the "directional dual norm"

$$\|F(z)\|_{*,\mathcal{Z},z} = \sup_{z' \in \mathcal{Z}} \frac{\langle F(z), z' - z \rangle}{\|z' - z\|}.$$

  Funnily enough, this gives rise to another min-max optimization: $\min_z \max_{z'} \frac{\langle F(z), z'-z \rangle}{\|z'-z\|}$, which looks even less tractable than the original one, but might be interesting to consider in some (?) cases if different oracles than the usual ones are available.

Remark 1.2. Aficionados of second-order methods will likely find JM a bit fishy. Indeed in the case $d_y = 0$, so that $\min_x \max_y f(x, y)$ just amounts to $\min_x f(x)$, the JM update reads

$$x^+ = x - \eta \nabla^2 f(x) \cdot \nabla f(x),$$

which is reminiscent of (damped) Newton's method

$$x^+ = x - \eta \left[\nabla^2 f(x)\right]^{-1} \nabla f(x),$$

but it uses the Hessian itself instead of Hessian inverse! So for $f(x) = \frac{1}{2} x^\top A x$ for example, Newton's method (with $\eta = 1$) converges in one step, so it's much nicer than gradient descent, whereas the JM update does something completely absurd. TODO: plot

Interestingly, it is indeed possible to use Newton [ZWPY21] or quasi-Newton methods [AL22] for min-max optimization, and it's just not the same method as JM.

2 Approximating the Jacobian method

Hessian-vector multiplications may be implementable efficiently, but it can still be a hassle. Here is a cheap variant that has the advantage of being a first-order method. It's just an application of the cheapest trick in the book: approximate gradients (in this case, Jacobians) by finite differences.

Fix $\eta, \delta > 0$ and assume F is C^2 (i.e. f is C^3). Then

$$\begin{aligned} F(z - \eta F(z)) - F(z) &= -\nabla F(z) \cdot \eta F(z) + O(\eta^2) \\ \frac{1}{\eta}\left(F(z - \eta F(z)) - F(z)\right) &= -\nabla F(z) \cdot F(z) + O(\eta) \\ &= -\frac{1}{2} \nabla \|F\|^2 (z) + O(\eta). \end{aligned}$$

So we can just iterate the update rule

$$\begin{aligned} z^+ &= z + \frac{\delta}{\eta}\left(F(z - \eta F(z)) - F(z)\right) \\ &= z - \frac{\delta}{2} \nabla \|F\|^2 (z) + O(\delta\eta) \end{aligned}$$

to obtain, in the limit of $\eta \to 0$, an approximation of JM. In the spirit of [Lu21], if we set $\delta \asymp \eta$, we get an order-1 approximation of the continuous-time dynamics $\frac{d}{dt} z = -\frac{1}{2} \nabla \|F\|^2 (z)$.

The above algorithm only requires two evaluations of F per iteration, which is the same as EG+. Due to the nature of the update, it is not possible to apply the same trick as for OGDA to further reduce to one evaluation. If we are ok with three evaluations per iteration, and if we further assume F is C^3 (i.e. f is C^4), just by symmetrizing we get one order closer to the desired dynamics:

$$\begin{aligned} F(z - \eta F(z)) - F(z) &= -\nabla F(z) \cdot \eta F(z) + \eta^2 F(z)^\top \nabla^2 F(z) F(z) + O(\eta^3) \\ \text{and} \quad F(z + \eta F(z)) - F(z) &= \nabla F(z) \cdot \eta F(z) + \eta^2 F(z)^\top \nabla^2 F(z) F(z) + O(\eta^3) \\ \text{so} \quad \frac{1}{\eta} \left(F(z - \eta F(z)) - F(z + \eta F(z)) \right) &= -2 \nabla F(z) \cdot F(z) + O(\eta^2) \end{aligned}$$

and so we can also just iterate

$$\begin{aligned} z^+ &= z + \frac{\delta}{2\eta} \left(F(z - \eta F(z)) - F(z + \eta F(z)) \right) \\ &= z - \frac{\delta}{2} \nabla \|F\|^2 (z) + O(\delta \eta^2). \end{aligned}$$

This allows us to take larger step-sizes.

3 Conclusion

I have many questions:

- How can the (approximate) Jacobian method be adapted to constrained problems? We discussed this already in the remark at the end of the first section.
- How to analyze convergence of the (approximate) Jacobian method? [ALW19] gives sufficient conditions for $z \mapsto \frac{1}{2} \|F(z)\|^2$ to satisfy the Polyak-Lojasiewicz (PL) inequality, but it would be interesting to go beyond that.
- Is there a relation between the PL condition on $\frac{1}{2} \|F\|^2$

  $$\forall z \in \mathcal{Z}, \ \frac{1}{2} \|\nabla F(z) \cdot F(z)\|^2 \geq \mu \frac{1}{2} \|F(z)\|^2$$

  and the weak Minty condition

  $$\exists z_0, \ \forall z \in \mathcal{Z}, \ \frac{\rho}{2} \|F(z)\|^2 \geq \langle F(z), z_0 - z \rangle \quad ?$$

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